

When Fine-Tuning Drifts: Mitigating Policy Drift with Base Action-Field Regularization

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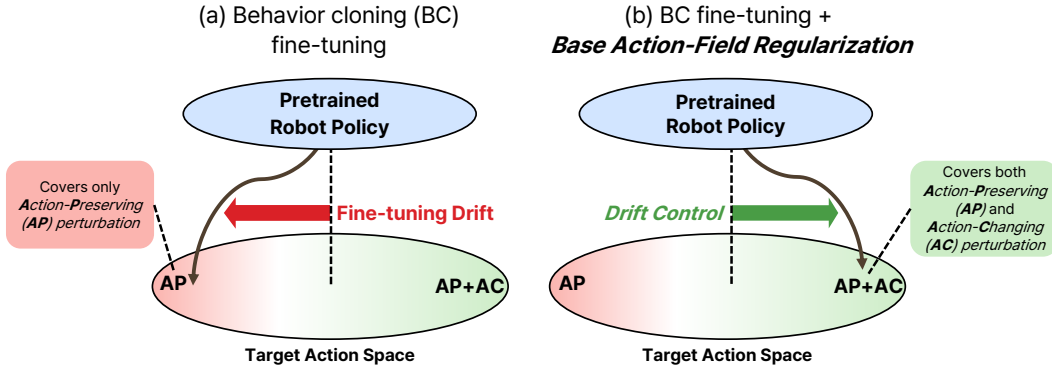


Figure 1: **Conceptual Illustration.** (a) In few-demonstration behavioral cloning (BC) fine-tuning, the policy can overfit to demonstrated target actions [1, 2], covering action-preserving (AP) cases where the pre-perturbation action remains sufficient while under-covering action-changing (AC) cases where a different action is required. (b) Base Action-Field Regularization controls this drift by preserving the pretrained action prior during adaptation, leading to broader coverage of both AP and AC regions in the target action space.

Abstract: Generalist robot policies acquire broad visuomotor priors through pre-training. However, few-demonstration behavior cloning (BC) fine-tuning can overfit to demonstrated action patterns while still appearing successful on the original task distribution. The resulting policy may appear successful when repeating demonstrated actions suffices, but becomes less robust when it should output a different action. We refer to this phenomenon as *fine-tuning drift*. To analyze fine-tuning drift, we introduce two quantitative diagnostics: *Target Error*, which quantifies the mismatch between target and predicted actions, and *Prior Drift*, which measures deviation from the pretrained action prior. We observe that Target Error and Prior Drift typically have a trade-off behavior, yet a robustly adapted policy should achieve both low Target Error and low Prior Drift. Building on this insight, we propose Target Error–Prior Drift Pareto control to select a robustly adapted policy from candidates. Furthermore, we propose *Base Action-Field Regularization* to incorporate this diagnosis in training. On the LIBERO-PRO benchmark, our method improves action-changing success rate from 26.12% to 42.12% while preserving action-preserving performance, improving the overall success rate from 65.70% to 73.05%. Code will be released upon acceptance.

1 Introduction

Generalist robot policies are increasingly employed as reusable foundations for downstream manipulation tasks. Pretraining on diverse datasets enables the acquisition of broad visuomotor priors that

are challenging to get from limited target data alone [3, 4, 5]. Vision-language-action models (VLAs) additionally introduce language-conditioned semantic priors that facilitate instruction-following control [6, 7, 8, 5]. Given that collecting extensive robot demonstrations is both costly and time-consuming [9], such prior knowledge is particularly valuable in few-demonstration scenarios, as it provides a good initialization for subsequent adaptation.

In practice, adaptation is typically achieved through behavior cloning (BC) fine-tuning, which involves collecting demonstrations, fine-tuning the policy, and evaluating performance on in-distribution manipulation tasks [8, 4]. While this approach often shows effectiveness, it can lead to overfitting the policy to demonstrated action patterns when data are limited [1, 2]. In such a case, the policy repeats the demonstrated action and succeeds when similar actions are sufficient to solve a target task. However, overfitting becomes particularly problematic when an adapted different action is required to deal with a slightly different task. For example, if the target position changes, the action should be changed accordingly.

We refer to this phenomenon as *BC fine-tuning drift* because BC-only fine-tuning improves in-distribution performance but reduces generalizability (Figure 1(a)). To diagnose fine-tuning drift, we introduce two quantitative diagnostics: Target Error, which measures the mismatch between target demonstration actions and the model’s predictions, and Prior Drift, which quantifies changes in action predictions relative to the initial pretrained policy. We observe that Target Error and Prior Drift typically exhibit a trade-off behavior. However, robust adaptation requires to achieve both low Target Error and low Prior Drift, so that the high generalizability is achieved. These findings hint that robust adaptation should not focus solely on minimizing Target Error.

Based on these observations, we propose a simple principle for preserving useful pretrained priors after or during fine-tuning. Specifically, we introduce **Target Error–Prior Drift Pareto control** as a model selection strategy. This approach identifies checkpoints that achieve favorable trade-offs between Target Error and Prior Drift from a Pareto-style perspective. In addition, we propose **Base Action-Field Regularization**, a training-time regularizer that explicitly constrains deviations from the pretrained action-field, thereby reducing unnecessary prior drift during adaptation (Figure 1(b)).

On LIBERO-PRO [10] with $\pi_{0.5}$ [5], Pareto drift control improves the action-changing success rate from 26.12% to 35.88%, and Base Action-Field Regularization further increases it to 42.12%, while preserving action-preserving performance and improving the overall success rate from 65.70% to 73.05%. These results show that controlling fine-tuning drift is most beneficial for the tasks requiring for the policy to change its action under task deviations, *i.e.*, perturbation. We summarize our main contributions as follows:

- **Fine-Tuning Drift Diagnosis.** We diagnose fine-tuning drift with action-preserving/action-changing evaluation axes and Target Error–Prior Drift scores, showing that lower target error can coincide with larger prior drift and weaker action-changing robustness.
- **Target Error–Prior Drift Pareto Control.** We use a Pareto-style diagnostic plane and shortest normalized-distance rule to select robust checkpoints without perturbation rollouts.
- **Base Action-Field Regularization.** Guided by the diagnosis, we add a training-time action-field regularizer that controls Prior Drift and further improves action-changing robustness.

2 Related Work

Pretrained Robot Policy and Adaptation. Pretrained robot policies are widely used as foundational models for robot adaptation. Research such as BC-Z [11] investigates zero-shot task generalization via language- and video-conditioned imitation. RT-1 [3] scales transformer-based policies using extensive real-world robot datasets, while RT-2 [7] incorporates web-pretrained vision-language models as vision-language-action (VLA) policies. Developments including Open X-Embodiment/RT-X [12], OpenVLA [8, 13], Octo [4], π_0 [14], and $\pi_{0.5}$ [5], have further established large-scale robot policies as reusable foundations that can be adapted to downstream tasks with limited demonstrations.

Most adaptation approaches rely on behavior cloning (BC) [15], which is trained on expert state-action pairs. However, BC policies may exploit correlations that are predictive in the training data but not causally relevant to task success, a phenomenon referred to as causal confusion [16]. This work investigates how fine-tuning a pretrained policy with a limited number of BC demonstrations can induce policy drift, thereby reducing the model’s ability to generalize to scenarios requiring appropriate action adaptation.

Perturbation-Based Robustness Evaluation. Models may fail when evaluated on challenging subpopulations or under distributional shifts [17, 18]. WILDS [19] formalizes the study of in-the-wild distribution shifts; shortcut learning [20] demonstrates that models may exploit predictive but non-causal cues that do not generalize under shifted conditions. In the context of robotic manipulation, LIBERO-PRO [10] introduces controlled perturbations to assess robustness beyond standard LIBERO [21] success. This work investigates few-demonstration fine-tuning and control drift prior to preserve the pretrained prior of VLA.

Prior-Preserving Adaptation. Preserving the pretrained prior during adaptation remains a significant challenge. Previous research has addressed this by constraining parameters [1, 22] or by aligning the behavior of earlier models [23, 24, 25]. An alternative strategy involves combining model parameters after fine-tuning [2, 26, 27]. For VLA, RETAIN [28] adopt parameter merging. This work introduces two strategies for model selection and regularization during training. In this work, we diagnoses action-field drift during fine-tuning and leverages this insight to guide Pareto drift control and fine-tuning regularization.

3 Method

In Sec. 3.1, we describe two quantitative values to study behavior cloning. Then, we introduce the methods for the model selection (Sec. 3.2) and the training (Sec. 3.3).

3.1 Diagnostic Axes for Drift-Controlled Adaptation

We view behavior cloning (BC) adaptation as balancing two competing effects: (1) fitting the target demonstrations, and (2) preserving the pretrained action-field. Let $\mathcal{D} = \{(o, a)\}$ denote the target demonstrations, where o is an observation and a is an expert action chunk. We adapt policy parameters θ from pretrained parameters θ_0 . While stronger fitting can improve matching to the target data, it may also induce larger deviation from the pretrained policy, potentially reducing robustness to states outside the target demonstration manifold. Conceptually, we characterize adaptation as:

$$\text{Robust Policy Adaptation} \approx \text{Target Fitting} + \text{Prior Preservation} \quad (1)$$

We define two diagnostic quantities to analyze adaptation behavior. We use a fixed diagnostic batch $\mathcal{B}_{\text{diag}}$ from the LIBERO training data, with 8 samples for each of the 40 tasks. First, we define **Target Error** $T(\theta)$ as the BC objective evaluated on $\mathcal{B}_{\text{diag}}$, measuring how well the adapted policy fits the target demonstrations. Second, we define **Prior Drift** $D(\theta)$ as the deviation between the adapted policy and the frozen pretrained policy given the same inputs, measuring how far the adapted policy moves from the pretrained action prior during adaptation.

Standard behavior cloning. In standard BC [15], the policy regresses expert actions from observations using an ℓ_2 objective. Accordingly, **Target Error** corresponds to the BC loss on the diagnostic batch, while **Prior Drift** measures the deviation from the frozen pretrained policy:

$$T(\theta) = \mathbb{E}_{(o,a) \sim \mathcal{B}_{\text{diag}}} [\|\pi_{\theta}(o) - a\|_2^2], \quad D(\theta) = \mathbb{E}_{(o,a) \sim \mathcal{B}_{\text{diag}}} [\|\pi_{\theta}(o) - \pi_{\theta_0}(o)\|_2^2]. \quad (2)$$

Flow-matching behavior cloning. Flow-matching formulations [29, 30] have been widely adopted for robot policy learning [14, 5]. Instead of directly regressing actions, the policy learns a velocity (action) field over noisy action states and diffusion time steps. Given Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$ and interpolation time $t \sim \mathcal{U}(0, 1)$, the noisy action state and target velocity are defined as

$$x_t = t\epsilon + (1 - t)a, \quad u_t = \epsilon - a. \quad (3)$$

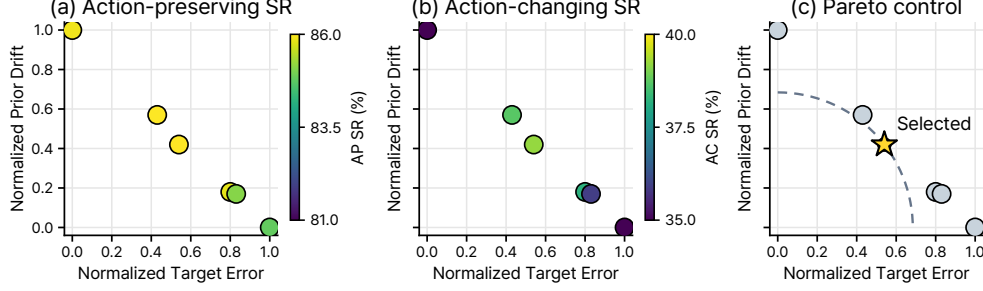


Figure 2: **Candidate selection on the normalized Target Error–Prior Drift plane.** AP and AC denote action-preserving and action-changing perturbations. We plot candidate policies as circles by varying batch size and training steps, using the same candidates across (a–c). Target Error and Prior Drift show an inverse trade-off: AP success remains consistently high across most candidate policies, while AC success peaks near a balanced point. The selected candidate in (c) is the first point reached by expanding a dashed circle from the lower-left corner.

The flow-matching policy $v_\theta(o, x_t, t)$ is trained to predict the target velocity. Accordingly, **Target Error** corresponds to the flow-matching objective evaluated on the diagnostic batch, while **Prior Drift** measures the deviation from the frozen pretrained velocity field:

$$T(\theta) = \mathbb{E}_{\mathcal{B}_{\text{diag}}, \epsilon, t} [\|v_\theta(o, x_t, t) - u_t\|_2^2], \quad D(\theta) = \mathbb{E}_{\mathcal{B}_{\text{diag}}, \epsilon, t} [\|v_\theta(o, x_t, t) - v_{\theta_0}(o, x_t, t)\|_2^2]. \quad (4)$$

Diagnostic use. **Target Error** and **Prior Drift** provide a two-dimensional view of adaptation. Low **Target Error** indicates strong target fitting, whereas low **Prior Drift** indicates preservation of the pretrained action-field. Robust policy adaptation should balance both, corresponding to candidates near the lower-left region of the Target Error–Prior Drift plane. We use this plane to diagnose fine-tuning drift, select Pareto-controlled checkpoints, and motivate the regularization of **Prior Drift**.

3.2 Diagnosing and Controlling Fine-Tuning Drift

We categorize perturbation scenarios by whether the original action remains same. **Action-Preserving (AP)** perturbations preserve the required action, such as changing object appearance or rephrasing the instruction while the same manipulation remains correct. **Action-Changing (AC)** perturbations require action changes, such as swapping the target object or changing the goal relation. In LIBERO-PRO [10], appearance/language perturbations are AP, while swap/task perturbations are AC.

We next empirically examine the relationship between the two diagnostic quantities introduced in Eq. 1. Based on the interpretation above, we expect **Target Error** and **Prior Drift** to exhibit a trade-off relationship during adaptation. To verify this tendency, we fine-tune policies under diverse training configurations by varying the batch size and training steps.

Figure 2 visualizes the resulting candidate policies using normalized **Target Error** on the x-axis and normalized **Prior Drift** on the y-axis. We observe three key trends:

- Figures 2(a,b) show a clear inverse relationship between **Target Error** and **Prior Drift**.
- As shown in Fig. 2(a), AP success remains consistently high across most candidate policies, indicating that same-action cases are already stable.
- In contrast, Figure 2(b) shows that AC success peaks near an intermediate trade-off region, suggesting that successful action adaptation requires both sufficient target fitting and preservation of the pretrained action-field.

These results suggest that minimizing target fitting error alone is insufficient for robust adaptation. Even for the tasks where action is required to be changed for solving task, the adapted policy must also preserve enough of the pretrained action-field.

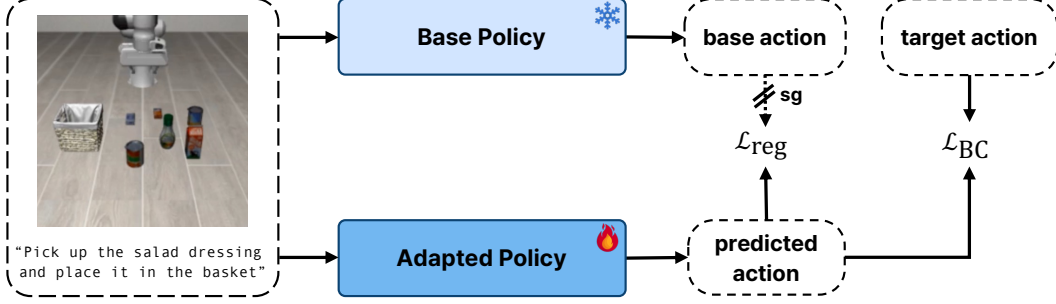


Figure 3: **Base Action-Field Regularization.** The adapted policy is trained with BC supervision, while a frozen base policy provides a stop-gradient action-field reference. The regularization term limits drift from the pretrained action-field during fine-tuning.

143 **Target Error–Prior Drift Pareto control.** When multiple fine-tuned policies are trained under
 144 varying configurations, a key challenge is to select a robustly adapted policy without depending
 145 on perturbation-based rollout evaluation. Minimizing only Target Error or only Prior Drift does
 146 not ensure optimal robustness to action changes. Robust adaptation is typically achieved near the
 147 Pareto-optimal trade-off region between target fitting and prior preservation.

148 Building on this observation, policies are selected using the Target Error–Prior Drift Pareto frontier.
 149 Let $\mathcal{A}$ represent the set of candidate policies, each evaluated using the two diagnostic quantities
 150 prior to perturbation rollout evaluation. We normalize T and D across $\mathcal{A}$ to obtain $\hat{T}$ and $\hat{D}$. With
 151 normalized scores $\hat{T}(\theta)$ and $\hat{D}(\theta)$, a balanced candidate is chosen such that both diagnostic quantities
 152 are jointly minimized:

$$r(\theta_i) = \sqrt{\hat{T}(\theta_i)^2 + \hat{D}(\theta_i)^2}, \quad \theta^* = \arg \min_{\theta_i \in \mathcal{A}} r(\theta_i). \quad (5)$$

153 This selection rule identifies a point on the Pareto frontier where target fitting and prior preservation
 154 are jointly balanced. Notably, the selection process depends exclusively on the proposed diagnostic
 155 quantities and does not incorporate perturbation rollout success labels.

156 3.3 Base Action-Field Regularization

157 The preceding Pareto-based policy selection shows that robust adaptation should control both **Target**
 158 **Error** and **Prior Drift**. Rather than only selecting among already fine-tuned candidates, we add the
 159 prior-drift control to the training objective. Candidate selection can require multiple training runs,
 160 while training-time regularization can encourage a similar balance within training procedure.

161 We express this goal as fitting the target demonstrations while limiting excessive drift from the
 162 pretrained action-field:

$$\min_{\theta} T(\theta) \quad \text{subject to} \quad D(\theta) \leq \delta, \quad (6)$$

163 Relaxing this constraint gives the following regularized objective:

$$\min_{\theta} T(\theta) + \lambda D(\theta), \quad (7)$$

164 where δ is the drift threshold and $\lambda \geq 0$ balances target fitting and prior preservation.

165 In standard behavior cloning, these two terms correspond to

$$T(\theta) \approx \mathcal{L}_{\text{BC}}(\theta), \quad D(\theta) \approx \mathcal{L}_{\text{reg}}(\theta, \theta_0), \quad (8)$$

166 where

$$\mathcal{L}_{\text{reg}}(\theta, \theta_0) = \mathbb{E}_{(o,a) \sim \mathcal{D}} \left[\|\pi_{\theta}(o) - \text{sg}(\pi_{\theta_0}(o))\|_2^2 \right]. \quad (9)$$

167 Similarly, in flow-matching behavior cloning, the prior-preservation term is given by

$$\mathcal{L}_{\text{reg}}(\theta, \theta_0) = \mathbb{E}_{(o,a) \sim \mathcal{D}, \epsilon, t} \left[\|v_{\theta}(o, x_t, t) - \text{sg}(v_{\theta_0}(o, x_t, t))\|_2^2 \right], \quad (10)$$

where $\text{sg}(\cdot)$ indicates that gradients are stopped through the pretrained policy. The resulting practical training objective is

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{BC}}(\theta) + \lambda_{\text{act}} \mathcal{L}_{\text{reg}}(\theta, \theta_0), \quad (11)$$

where λ_{act} is the regularization coefficient. This objective fits the target demonstrations while limiting excessive drift from the pretrained action-field during adaptation.

4 Experiments

We first describe the experimental setup (Sec. 4.1), then report results on original and perturbed scenarios (Sec. 4.2). Finally, we analyze the proposed method (Sec. 4.3).

4.1 Experimental Setup

Models. We use $\pi_{0.5}$ [5] as our primary backbone and additionally evaluate OpenVLA-OFT [13]. All models are initialized from the official checkpoints released by the corresponding repositories.

Benchmarks. We fine-tune on target demonstrations from LIBERO [21] and evaluate robustness on LIBERO-PRO [10]. We report original LIBERO success and four perturbation axes: object perturbations modify object appearance, language perturbations rephrase the instruction, swap perturbations change the target object, and task perturbations alter the goal relation or required manipulation. Object and language perturbations are treated as action-preserving because the correct behavior can remain the same, while swap and task perturbations are action-changing because the policy should change its action when the target object, goal relation, or required manipulation changes. We use rollout success rate as the metric, following the official LIBERO-PRO evaluation protocol.

Implementation details. For Pareto drift control, candidates are generated by varying standard fine-tuning hyperparameters such as batch size and training steps, and the candidate closest to the origin in the normalized Target Error–Prior Drift plane is selected using diagnostic scores only. In our primary $\pi_{0.5}$ setting, this selects a batch size of 64 and the 25k-step checkpoint. For Base Action-Field Regularization, we start a separate fine-tuning run from the same pretrained initialization and select the regularization coefficient λ_{act} using the same Pareto diagnostic criterion.

4.2 Action-Preserving and Action-Changing Robustness

Drift-control effectiveness. We evaluate whether controlling fine-tuning drift improves success rates under original and perturbed LIBERO-PRO scenarios. Table 1 reports performance on original tasks, action-preserving perturbations, and action-changing perturbations. The $\pi_{0.5}$ maintains high action-preserving success, with an AP average of 91.50%, but drops substantially on action-changing perturbations, with an AC average of 26.12%. Pareto drift control improves the AC average to 35.88%, and Base Action-Field Regularization further raises it to 42.12%, while preserving the AP average at 91.50%. The overall average also increases from 65.70% to 73.05%. Since the full method uses the same selected training configuration and only adds the action-field coefficient, this result suggests that training-time Prior Drift control complements diagnostic candidate selection. Overall, the gains are concentrated on action-changing perturbations, supporting our claim that fine-tuning drift is most harmful when the policy must modify its action.

Comparison with recent robustness strategies. We next compare with related robustness strategies that intervene at different stages of the policy. Vision-Language Steering (VLS) [31] modifies a frozen policy at test time, while RETAIN [28] performs post-hoc parameter merging between the pretrained checkpoint and the fine-tuned checkpoint. For RETAIN, we report the best α selected by the average success over swap and task perturbations. Even with this favorable selection, RETAIN yields only a marginal improvement over the $\pi_{0.5}$ on action-changing perturbations. In contrast, drift-controlled fine-tuning substantially improves the average success on the same axes, with the full method reaching 42.12%. These results suggest that directly controlling action-field drift during adaptation provides a strong robustness strategy for action-changing perturbations.

Table 1: **Rollout success rate (%) on LIBERO and LIBERO-PRO.** Object and language perturbations are action-preserving, while swap and task perturbations are action-changing. AP Avg. averages object and language success; AC Avg. averages swap and task success; All Avg. averages original success and all perturbation axes.

Method	Orig	Action Preserving			Action Changing			All Avg.
		Obj	Lang	Avg.	Swap	Task	Avg.	
$\pi_{0.5}$	93.25	86.25	96.75	91.50	27.00	25.25	26.12	65.70
+ Pareto drift control	95.50	81.00	97.00	89.00	40.00	31.75	35.88	69.05
+ Base Action-Field Reg.	98.00	84.75	98.25	91.50	44.75	39.50	42.12	73.05

Table 2: **Comparison with related robustness strategies in success rate (%).** The table provides a compact comparison on action-changing axes; methods differ in intervention stage. RETAIN merges the pretrained and fine-tuned checkpoints, and we report the best α selected by average success on these two perturbation axes.

Method	Strategy	Swap pert.	Task pert.	Avg. SR
$\pi_{0.5}$	None	27.00	25.25	26.12
+ VLS (arXiv’26) [31]	Test-time steering	35.13	38.50	36.82
Best RETAIN (ICLR’26, $\alpha = 0.9$) [28]	Weight interpolation	29.30	25.10	27.20
+ Pareto drift control	Pareto selection	40.00	31.75	35.88
+ Base Action-Field Reg.	Regularized fine-tuning	44.75	39.50	42.12

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Backbone generalization. To examine whether drift control is specific to $\pi_{0.5}$, we further apply our method to OpenVLA-OFT [13]. Table 3 reports an additional backbone check on task perturbations, which require action changes under perturbation. Improvement with our method provides initial evidence that the proposed drift-control principle transfers beyond the primary backbone. OpenVLA-OFT details are provided in the supplement.

Table 3: **OpenVLA-OFT task-perturbation success rate (%).**

Method	Task Pert.
OpenVLA-OFT [13]	9.25
+ Ours	19.30

214 4.3 Qualitative Analysis

Perturbation landscapes. We first provide a qualitative view of how drift control influences performance across the perturbation space. Figure 4 presents an interpolated success-rate landscape over action-preserving and action-changing perturbation coordinates. As the approach transitions from $\pi_{0.5}$ to Pareto drift control and subsequently to Base Action-Field Regularization, the high-success region expands, particularly along the action-changing axis, as indicated by the red arrow in Fig. 4. This pattern aligns with the rollout results in Table 1. Furthermore, this finding supports the intuition depicted in Figure 1: drift control improves performance under perturbations.

Action retargeting examples. We next inspect rollout examples on action-changing perturbations. Figure 5 shows that $\pi_{0.5}$ often continues to follow the action pattern associated with the pre-perturbation setting even after the target object or goal relation changes. In contrast, our drift-controlled policy does not simply repeat the original behavior, but adjusts its action to the perturbed target and successfully completes the manipulation. These examples illustrate the behavior difference behind the quantitative gains in swap and task success.

228 5 Limitations

Our diagnostic scores are not a substitute for rollout evaluation and do not certify OOD robustness. They are used to select adaptation candidates before LIBERO-PRO evaluation, while final claims are still based on rollout success. Our main experiments focus on simulation benchmarks with $\pi_{0.5}$, with OpenVLA-OFT included as an additional backbone. This controlled setting allows us

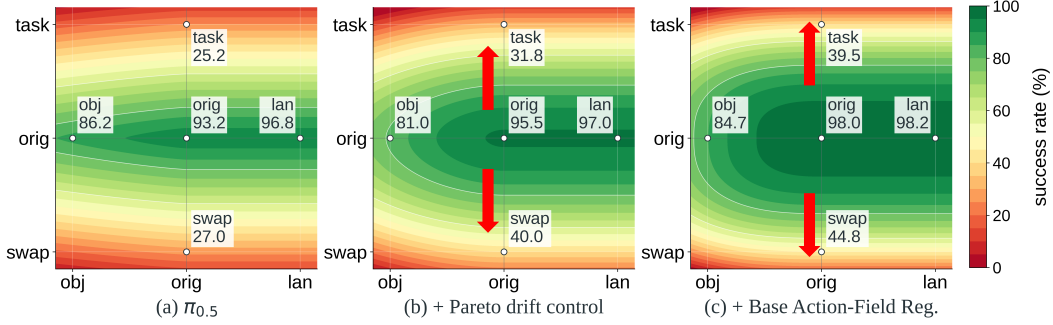


Figure 4: **Action-perturbation performance landscapes.** We visualize interpolated landscapes over action-preserving and action-changing perturbation coordinates for $\pi_{0.5}$, Pareto drift control, and Base Action-Field Regularization. From left to right, the high-success region expands, especially along the action-changing axis, consistent with improved robustness under swap and task perturbations.

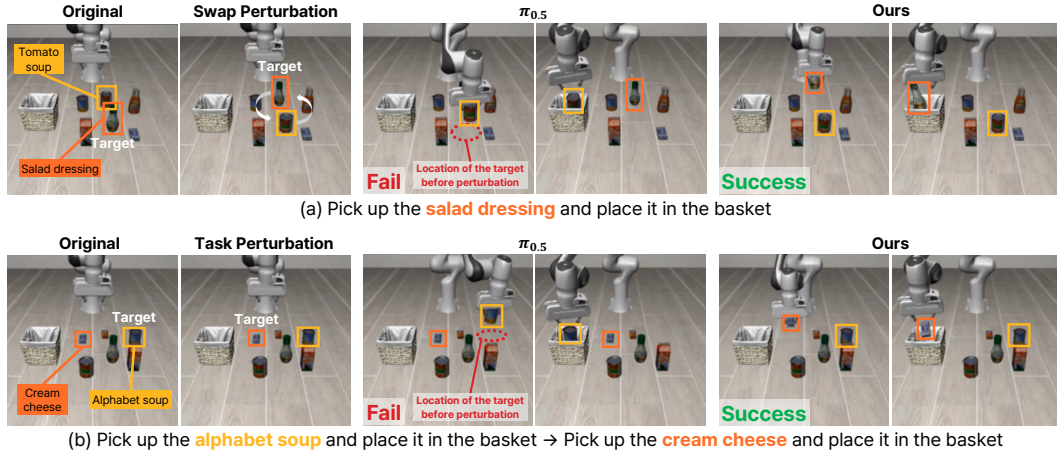


Figure 5: **Action retargeting under action-changing perturbations.** We compare $\pi_{0.5}$ with our drift-controlled policy. While $\pi_{0.5}$ often repeats the pre-perturbation action pattern, our policy adapts more consistently to the perturbed target and completes the manipulation.

233 to isolate fine-tuning drift and evaluate action-preserving versus action-changing robustness under
 234 consistent perturbation protocols. Extending the diagnostic to broader task families and perturbation
 235 types would further clarify when prior-drift control is most useful. Broader validation across more
 236 generalist policies remains an important direction for future work.

237 6 Conclusion

238 We study the adaptation of generalist robot policies using few target demonstrations. We introduce
 239 two diagnostics: Target Error, which quantifies agreement with target demonstrations, and Prior Drift,
 240 which assesses deviation from the pre-trained policy prior. The analysis demonstrates a consistent
 241 trade-off between these metrics, as reducing Target Error frequently increases Prior Drift. This
 242 finding indicates that adaptation may overfit to target demonstration performance while diminishing
 243 behaviors essential for generalization. To address this challenge, two methods for controlling policy
 244 drift are proposed: Target Error–Prior Drift Pareto Control, a model selection strategy that identifies
 245 checkpoints with both low target error and low prior drift, and Base Action-Field Regularization, a
 246 training-time objective that preserves the pre-trained action prior during adaptation. Evaluations on
 247 LIBERO-PRO show that both methods consistently enhance robustness and generalization compared
 248 to existing baselines. These findings indicate that effective adaptation requires not only fitting target
 249 demonstrations but also maintaining the prior from the pre-trained VLA, and that explicitly measuring
 250 and controlling policy drift offers a practical approach to achieving more reliable robot adaptation.

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